

Beyond Word Error Rate: Task-Success Benchmarking of Code-Switched Twi–English and Pidgin–English Speech for a Clinical Agent

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Abstract

Speech recognition for African languages is normally reported as word error rate (WER). In a deployed clinical agent, what matters is not the transcript but the action it produces. We benchmark code-switched Ghanaian clinical speech — Twi–English and Pidgin–English — on downstream task success: blood-pressure extraction, escalation correctness under a fixed clinical rule, and intent. The harness imports the production speech layer of a deployed WhatsApp chronic-care agent, so the systems measured are the systems serving patients. Across an African-built API (Intron Sahara, in two language-hint configurations), a commercial API (Cartesia Ink) and an offline open-weights model (faster-whisper), we find that *the language hint given to a model changes downstream accuracy more than the choice of model does*, and that its effect is opposite in the two Ghanaian languages: the pcm hint makes Sahara the strongest system on Pidgin–English by a wide margin (WER 0.085, intent 93%), while the tw hint returns empty transcripts on 16 of 27 Twi–English utterances and drives escalation correctness to zero. Pinning the same API to English instead recovers 78% on Twi–English. We report escalation failures by direction, since a missed escalation and a false alarm are not equivalent harms.

Index Terms: speech recognition, code-switching, low-resource languages, Twi, Ghanaian Pidgin, task-based evaluation, digital health

1 Introduction

In a deployed clinical agent no clinician reads the transcript. What reaches a human is an *action*: a blood-pressure reading recorded against a patient, an escalation raised to a nurse, a refill routed to a pharmacy. A benchmark that stops at WER measures the intermediate artefact, not the thing the system is for.

This matters acutely in West African clinical speech, which is routinely code-switched. Ghanaian patients mix Twi or Ghanaian Pidgin with English, and the clinically load-bearing tokens — numerals and drug names — are almost always spoken in English inside an otherwise non-English utterance. A model can therefore mis-transcribe most of a sentence and still produce the correct action, or transcribe most of it correctly and destroy the action.

We contribute a task-success harness that scores blood-pressure extraction, escalation correctness and intent alongside WER and CER, and which *imports the production speech layer* rather than reimplementing it; a consented, de-identified corpus of Ghanaian code-switched clinical utterances; and a four-configuration comparison spanning an African-built API, a commercial API and an open-weights model running fully offline on CPU.

2 System and task

VeloxaCare receives patient messages over WhatsApp as text or voice notes. A speech layer transcribes voice; a rule-based component then extracts structured content and decides what happens next.

Escalation is deliberately *not* model-driven. A systolic reading ≥ 160 or diastolic ≥ 100 escalates immediately; a reported cost barrier or side effect escalates after two occurrences within fourteen days. Language models structure,

classify and summarise; they never diagnose and never decide escalation. That boundary is a medical-safety requirement, and it is what makes this benchmark meaningful: because the decision rule is fixed and identical across systems, any difference in escalation outcome is attributable to the transcript alone.

The corpus contains a deliberate minimal pair. ``Me BP yε one-sixteen over seventy-eight'' is 116/78 and green; "one-sixty over one hundred" is 160/100 and red. One vowel separates a routine reading from an immediate escalation.

3 Data

Source. Twenty-one scripted scenarios, written to cover the intents the agent actually handles: adherence confirmation, cost barrier, forgetting, side effect, running out, blood-pressure reporting, booking, rescheduling and refills. The scenario file is the single source of truth from which the scoring reference is generated; hand-editing the reference is prohibited, because text that drifts from what speakers read makes every WER figure meaningless.

Languages. Three sets: E, five Ghanaian-accented English controls; T, nine Twi–English; P, five Pidgin–English. Numerals and drug names remain in English in every set — this is what Ghanaian speakers do, and it is precisely the code-switch boundary under test. Drug names (*amlodipine*, *lisinopril*, *nifedipine*) and blood-pressure values therefore sit inside Twi or Pidgin matrix sentences.

Size. Three consented speakers, recorded on personal handsets, giving 57 recordings totalling 5.3 minutes (0.088 h; mean utterance 5.6 s), scored across four configurations for 228 transcriptions. Coverage is uneven — 28, 19 and 10 recordings per speaker — and splits 15 English, 27 Twi–English, 15 Pidgin–English; 48 quiet and

9 with ambient noise. Three calls failed with network `ConnectionErrors` and are excluded from their aggregates. This is a small corpus and no absolute figure here should be read as a general claim about Ghanaian speech; what it supports is a controlled comparison, since every configuration receives identical audio, preprocessing, normalisation, scoring code and decision rules.

Preprocessing. Audio arrived in mixed encodings — some handsets produce Opus in an .m4a container, others AAC — and one API rejects the former outright. All audio is transcoded to 16kHz mono AAC before scoring so every system receives identical input. We report this because an earlier run failed on every recording from two speakers for exactly this reason and would, uninspected, have been written up as catastrophic model failure rather than a container mismatch. Reference and hypothesis text are normalised identically before scoring: lower-cased, punctuation stripped, number words converted to digits, and blood-pressure notation canonicalised so that 142/95 and “142 over 95” collapse to the same form.

4 Models

Intron Sahara is an African-built ASR API whose documented input languages include Twi (tw), Akan (ak), Ghanaian Pidgin (pcm) and Ga (gaa). We evaluate two configurations: **sahara**, sending the set’s non-English code, and **sahara_en**, pinning the English model regardless of what is spoken.

Cartesia Ink (ink-whisper) is a latency-optimised commercial ASR service, accessed over the network with an API key. **faster-whisper** (base) is the only system here that is not a hosted service: it runs Whisper’s openly released weights locally on CPU, with no key and no network.

Probing each endpoint with every language code, rather than trusting documentation, produced the cleanest result in the study. Cartesia returns an explicit HTTP 400: `invalid language` for tw, ak, pcm and gaa while accepting en and sw: the only African language it takes is Swahili. Whisper’s languages contain none of the four. Of the systems tested, only the African-built one can be told what language a Ghanaian patient is speaking.

Documented is not the same as shipped, and we state this plainly. Intron confirmed that the *Akan-English code-switch pair* — the configuration these utterances actually require — had not shipped at the time of testing, and was rolling out in the week of 10 August 2026.

5 Evaluation metrics

We report **WER** and **CER**. CER matters here because code-switched transcripts are frequently right about content and wrong about word boundaries; WER charges a full error for a token split that CER shows to be a near miss.

We then score what the agent does with the transcript. **BP extraction accuracy** asks whether the systolic/diastolic pair recovered matches the reference. **Escalation correctness** — our headline — asks whether the transcript drives the same green/amber/red decision under the fixed clinical rule; it is the only metric corresponding to a patient outcome. **Intent accuracy** asks whether the agent takes the

Table 1: Overall, 57 recordings, three speakers.

Model	WER	CER	BP	Esc.	Intent
Sahara (tw/pcm)	0.506	0.456	33%	39%	51%
Sahara (en)	0.415	0.240	67%	83%	75%
Cartesia Ink	0.455	0.257	61%	67%	75%
faster-whisper	0.589	0.356	56%	61%	61%

Table 2: WER / CER per language set.

Model	English		Twi-En		Pidgin-En	
	WER	CER	WER	CER	WER	CER
Sahara (tw/pcm)	0.130	0.045	0.949	0.887	0.085	0.066
Sahara (en)	0.130	0.045	0.607	0.364	0.354	0.197
Cartesia Ink	0.071	0.041	0.647	0.378	0.493	0.256
faster-whisper	0.139	0.080	0.791	0.476	0.676	0.418

right action. The intent scorer is rule-based and identical for every system, so it is a fair comparison even where it is an imperfect absolute measure.

These metrics are appropriate here because the system is autonomous between the patient and the nurse. No human reads the transcript before an escalation is raised or withheld, so transcript-level metrics alone cannot establish whether the system is safe.

Latency is not reported. A second benchmark run shared Sahara’s 30 requests/minute limit during this execution, so per-call timings measure contention rather than model speed; publishing them would mislead.

6 Results

The English control. Three of four configurations reach 100% escalation correctness on Ghanaian-accented English (Table ??); the offline model reaches 67%. Because the hosted systems are equivalent here, their degradation on the code-switched sets is attributable to code-switching rather than to accent, recording conditions or scoring code.

Code-switch penalty (mean WER on code-switched audio minus WER on the English control): +0.387 for Sahara (en), +0.510 for Sahara (tw/pcm), +0.521 for Cartesia, +0.611 for faster-whisper. Every system degrades; the African-built model under an English hint degrades least.

Noise. Sahara is markedly the most robust. Its WER is essentially unchanged from quiet to noisy (0.506 to 0.504; 0.417 to 0.405 under the English hint), while Cartesia degrades from 0.406 to 0.717 and faster-whisper from 0.557 to 0.760. For a clinic waiting room this matters more than a headline average.

Escalation failures by direction. Aggregate accuracy hides a clinically important asymmetry. Across 57 utterances, Sahara (tw/pcm) produced 11 escalation flips, **five of them missed escalations** — a reference-red reading that did not escalate — and no false alarms. Sahara (en) produced 3 flips (1 missed, 1 false alarm), Cartesia 5 flips (0 missed, 1 false alarm) and faster-whisper 7 flips (2 missed). These are not equivalent harms: a false alarm costs a nurse a phone call; a missed escalation is a reading of 160/100 or above that never reaches anyone.

Table 3: Downstream task success per language set, as BP extraction / escalation correctness / intent (%).

Model	English	Twi-En	Pidgin-En
Sahara (tw/pcm)	67/100/93	0/0/4	67/67/93
Sahara (en)	67/100/93	56/78/78	83/83/53
Cartesia Ink	100/100/100	56/67/78	50/50/47
faster-whisper	67/67/93	56/56/59	50/67/33

7 Qualitative findings

Sahara (tw/pcm): excellent on Pidgin, silent on Twi. The aggregate hides two opposite behaviours. With the pcm hint this is the best system in the study on Pidgin-English, and the only one that preserves Pidgin rather than Anglicising it:

REF “I no fit buy the medicine, e cost too much for me this month.”
SAHARA(PCM) “I no fit buy the medicine, e cost too much for me.”
CARTESIA “I know if you buy the medicine, it costs two months for me...”
F-WHISPER “I don’t feel bad about medicine, it costs too much...”

Cartesia’s rendering inverts the meaning: “I no fit buy” (I cannot afford) becomes “I know if you buy”. The cost barrier — the single most important signal this product detects — is destroyed. Sahara keeps it, and scores 93% intent on Pidgin against Cartesia’s 47%.

With the tw hint the same API returns an **empty transcript on 16 of 27 Twi-English utterances**, and on 0 of 15 Pidgin ones. It is not mishearing the English fragments; it does not emit them. Since those fragments carry the numerals and drug names, every downstream metric collapses: 0% BP, 0% escalation, 4% intent.

Sahara (en): the best all-round configuration. Pinning English recovers Twi-English to 78% escalation correctness and gives the best overall CER (0.240) and escalation (83%). It pays for this on Pidgin intent (53% against 93% under pcm), because the English model normalises Pidgin into standard English and the rule-based intent detector loses its keywords.

Cartesia Ink: strongest on English, weakest on Pidgin. Best English WER (0.071) and no missed escalations, but it cannot be told a Ghanaian language exists, and on Pidgin it falls to 50% BP and 47% intent. It also degrades most under noise. A representative failure: “I check my BP for house” became “I checked my BB for halves”.

faster-whisper: weakest, but free and offline. Highest WER and CER in every set, and it drops digits — on one Pidgin utterance it rendered 138/89 as “38 over 89”, silently losing the leading digit of a systolic reading. It is nonetheless the only configuration that runs with no key, no network and no cost.

The minimal pair. On the 116/78 utterance no system recovered both numbers; on the 160/100 utterance three of four recovered it exactly. The green reading is the harder one, and errors there manufacture escalations rather than suppress them.

Table 4: Summary of strengths and weaknesses.

Model	Strengths	Weaknesses
Sahara (tw/pcm)	Best WER/CER on intent of any system; preserves Pidgin rather than Anglicising it; noise-robust; API accepting Ghanaian codes	Pidgin tw returns empty transcripts on 16/27 Twi utterances; most missed escalations (5); Akan-English pair not yet shipped
Sahara (en)	Best overall CER and escalation (83%); small-code-switch penalty; noise-robust	Anglicises Pidgin, costing intent accuracy (53%); one missed escalation and one false alarm
Cartesia Ink	Best WER/CER; missed escalations	English no Ghanaian codes; inverts Pidgin meaning; worst noise degradation; weakest Pidgin task success
faster-whisper	No key, no network, no cost; fully offline; viable fallback	Weakest on every metric; drops digits in BP readings; only system below 100% on the English control

8 Discussion

The language hint dominates the model choice — in both directions. The largest effect in this study is not which system was used but what it was told, and the sign of that effect flips between the two Ghanaian languages. The pcm hint takes Sahara from mid-table to best-in-study on Pidgin. The tw hint takes the same API to zero on Twi. A benchmark that tested one hint, or averaged the two languages together, would have reported neither.

This reframes the result for practitioners: the right deployment is not one model but a per-language routing policy, and choosing it requires exactly the per-language task-success measurement reported here.

WER and task success disagree in both directions. In an earlier run Sahara scored worst on WER until we inspected the transcripts: it had heard “one forty-two over ninety-five” and written 142/95, recognising a blood pressure, while others wrote 142 over 95. Our normaliser counted one token against three, and WER punished the system that had understood best; after canonicalising BP notation it moved from worst to best, with nothing about the audio or models changed. The converse also occurs: we observed “medicine” transcribed as “men’s son” — badly wrong by any string metric — while the cost intent was still detected and the patient routed correctly. CER exposes part of the gap that WER hides, which is why we report both.

Offline operation degrades but does not collapse. The open-weights configuration is weakest on every metric, yet

still recovers the correct escalation on 61% of utterances with no key, no network and no cost. That is not good enough to be the primary path, but it is the right floor: when a clinic’s connection drops, the agent keeps making correct decisions on most messages instead of none.

9 Ethics, inclusion and limitations

All utterances are scripted; no real patient data was used. Written consent was obtained from every speaker before recording, with a separate opt-in for releasing audio. Speakers are identified only by code, never by name. Voice notes in the production system are stored under generated filenames.

The corpus is small — three speakers, 5.3 minutes — and unevenly distributed (28/19/10 recordings), so the aggregate is weighted towards one speaker and speaker effects cannot be fully separated from system effects; the escalation-flip counts rest on modest denominators. The rule-based intent scorer is tuned to these scenarios and flatters every system equally; it measures relative performance, not absolute agent quality. The Twi scripts were drafted and revised by non-native speakers and carry outstanding review questions, so native-speaker sign-off is required before these figures are cited as a general claim about Twi ASR. We would rather state these limits than have a reader find them.

10 Conclusion

Three results carry beyond this corpus. The language hint given to a model changed downstream accuracy more than the choice of model did, and in opposite directions for Twi and Pidgin — so the deployable answer is a per-language routing policy, not a single winner. WER can rank the most clinically useful transcript last and can penalise errors that change nothing, while escalation correctness was never fooled. And escalation errors must be reported by direction, because a missed escalation and a false alarm are not equivalent harms.

The practical implication is that code-switching must be evaluated explicitly. A system that performs well on monolingual Twi and well on monolingual English can still fail on the utterances patients actually produce — and fail in the direction that matters, by dropping the numbers.